



C23-EE-305

23138

BOARD DIPLOMA EXAMINATION, (C-23)

MARCH/APRIL—2025

DEEE – THIRD SEMESTER EXAMINATION

PROGRAMMING IN C

Time : 3 hours]

[Total Marks : 80

PART—A

3×10=30

- Instructions :**
- (1) Answer **all** questions.
 - (2) Each question carries **three** marks.
 - (3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

- 1. What is character set?
- 2. List various data types with examples.
- 3. What is conditional expression? Explain with simple example.
- 4. What is the purpose of break and continue statements?
- 5. Define two dimensional array and explain how it is declared and initialized.
- 6. Write a program in C to find the length of given string.
- 7. Define function and its use.
- 8. Differentiate between local and global variables.
- 9. Define structure.
- 10. Write a program in C to add two numbers using pointers.

PART—B

10×5=50

- Instructions :**
- (1) Answer *any five* questions.
 - (2) Each question carries **ten** marks.
 - (3) Answers should be comprehensive and the criteria for valuation is the content but not the length of the answer.

- 11. Explain in detail input/output functions in C with examples.
- 12. Describe various decision making statements in C. Provide examples to explain how each one is used.
- 13. Discuss in detail the concept of nesting of loops in C. Write a program to print the factorial of a given number.
- 14. Write a program in C to add two 3X3 matrices using arrays.
- 15. Explain various string handling functions with examples.
- 16. Write a program in C to add three numbers using the concept of functions.
- 17. Define union in C. How does a union differ from a structure in terms of memory allocation and usage? Provide an example.
- 18. Write a program to add marks of five subjects of each student with name, pin and marks using structures.

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